Nannymud — Game Guide

> A browser-native 2.5D beat-'em-up in the spirit of Little Fighter 2. Fifteen guilds, nine stages, three modes, one deterministic engine, zero audio files.

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What Is Nannymud?

Nannymud is a browser-native 2.5D side-scrolling beat-'em-up built over a sustained AI-assisted development sprint. It draws directly from Little Fighter 2 for its movement model — depth-plane dodging, jump arcs, directional combos, grab-and-throw — and layers on fifteen fully differentiated guild fighters, nine stylised stages, procedural audio, and server-authoritative online multiplayer.

Everything in the game was built from scratch with no external game assets:

- **UI** — designed with Claude's frontend skill
- **Lore** — written to fit the world of Nannymud

- **Game design** — adapted from the Little Fighter 2 model
- **Sprites** — AI-generated via PixelLab Pro (80px, displayed at ×1.5 scale)
- **Audio** — 100% procedurally synthesised via the Web Audio API
- **Multiplayer** — server-authoritative via Colyseus, WebSocket state sync
- **Simulation** — deterministic pure-TypeScript engine, shared client/server

Modes

Story Mode

Single-player run across the Plains of Nan. Six waves of escalating enemies culminate in a three-phase boss fight against the **Bandit King**. The stage scrolls right as you clear each wave; enemies spawn ahead of your position. Die once and it is over.

VS (1v1 vs CPU)

Pick any two guilds, pick a stage, fight. Best-of-three rounds. The AI opponent runs the same simulation logic as the server, at max difficulty. Guild matchups matter — there is a complete win-rate matrix to prove it.

Multiplayer (Online 1v1)

Server-authoritative match via Colyseus. Host creates a room, shares the code; guest joins. Both players run through the full lobby flow — character select, stage select — before the match starts. Best-of-three rounds. The server owns all simulation state; clients send inputs only.

Controls

Action	Default Key
Move left / right	← →
Move up / down (depth plane)	↑ ↓
Jump	Space
Run	Double-tap ← or →

Action	Default Key
Attack	J
Block	K
Grab / Throw	L
Pause	P
Fullscreen	F

Combo Grammar

Abilities are executed through directional input sequences followed by the attack button — no separate ability buttons.

Notation	Input
`↓↓J`	Down, down, attack
`→→J`	Right, right, attack
`↓↑J`	Down, up, attack
`↔J`	Left, right, attack
`↓↑↓↑J`	Down, up, down, up, attack (ultimate)
`K+J`	Hold block, press attack (RMB utility)

All keybinds are remappable in Settings and persist via localStorage.

The Guilds

Fifteen playable guilds, each with a distinct resource system, stat profile, and ability set covering every combat archetype in the game.

Stat Profiles

Guild	HP	Speed	STR	DEX	CON	INT	WIS	Armor	MR	Resource
Adventurer	520	140	14	12	14	8	10	10	5	Stamina

Guild	HP	Speed	STR	DEX	CON	INT	WIS	Armor	MR	Resource
Knight	580	120	14	8	18	8	12	12	10	Resolve
Mage	490	130	6	10	8	20	14	10	20	Mana
Druid	440	130	8	10	14	14	18	8	12	Essence
Hunter	450	140	10	18	12	8	10	6	5	Focus
Monk	460	160	10	18	12	8	14	7	10	Chi
Viking	460	125	18	10	16	6	8	8	5	Rage
Prophet	500	125	8	10	12	12	18	6	20	Faith
Vampire	530	145	10	16	12	10	8	8	10	Bloodpool
Cultist	300	125	6	10	10	18	10	4	12	Sanity
Champion	460	135	18	12	14	6	6	12	5	Bloodtally
Darkmage	300	125	6	10	10	18	14	4	14	Mana
Chef	540	130	10	10	12	14	10	6	6	Stamina
Leper	480	120	14	10	18	8	8	7	10	Rot
Master	620	135	12	12	14	14	14	10	10	Mastery

Guild Lore

Adventurers Guild — "The first guild a recruit walks into. No creed, no god — just iron in the hand and road in the boots. Adventurers fill every gap in every line."

Assembly of Knights — "Oath-sworn, plate-clad, sanctified. The Assembly holds the line when the line should not hold."

Mages Guild — "The tower accepts no half-measures. Frost, arcane, annihilation — cast clean or burn out."

Druids — "Keeper of the old groves. Shape becomes thought becomes shape again — wolf on the press, wolf on the hunt."

Hunters Guild — "Patient, precise, paired. The bow speaks, the wolf answers."

Holy Monks Order — "Breath before blow. Blow before breath. The Order teaches both; the student learns neither until they are the same."

Vikings Guild — "Raise the horn, make the tide red. The Vikings guild prefers its arguments concluded."

Prophets — "Reads the wind for signs. Paints the battle with blessings and the enemy with curses."

Vampires — "Coven of the long night. Moves where the lamps do not. Feeds, forgets, feeds again."

Cult of the Drowned — "They say the deep has eyes, and the eyes have names, and the names are vowels you should not make."

Champions of the Red Throne — "Blood for the throne. Never retreat — the throne counts your steps, and the wrong ones bleed."

Dark Guild — "The Dark Guild turned the lamps down and the books dark. What is learned here is not unlearned."

Chefs Guild — "An army marches on its stomach. The Chefs Guild argues this is in fact the only way anything marches."

Lepers — "Cast from the cities, kept from the temples, welcomed only by the rot. The Lepers return the favor."

Masters of Nannymud — "Few earn the title. Fewer keep it. The Masters don't choose a class — they rotate through yours."

Abilities & Combo Grammar

Ability Archetypes in Nannymud

The game contains one of the widest ability type spreads in a game of this scope:

Type	Guilds
Melee Specials	Adventurer, Knight, Monk, Viking, Champion, Leper, Druid (Wolf)
Ranged Projectiles	Mage, Hunter, Prophet, Vampire, Cultist, Darkmage, Adventurer, Viking

Type	Guilds
AoE Zones	Mage (Meteor), Druid (Wild Growth), Hunter (Rain of Arrows), Darkmage (Eternal Night), Leper (Rotting Tide), Cultist (Open the Gate)
Traps	Hunter (Bear Trap — root on trigger), Cultist (Tendril Grasp — ground root + DoT), Darkmage (Darkness — vision zone)
Summons / Pets	Hunter (Wolf companion, commandable AI), Cultist (Drowned Spawn, 150 HP), Leper (Rotting Husks from kills, 5 s)
Shapeshift	Druid (toggles Wolf form — fully separate ability set)
Ability Steal / Rotation	Master (Eclipse cycles all five primed-class ability variants; Class Swap RMB changes primed class)
Lifesteal	Vampire (Fang Strike heals 60% of damage; Blood Drain heals full damage; Hemorrhage overflows to heal), Viking (Whirlwind 10% lifesteal; Bloodlust 10% lifesteal)
DoT / Infection	Cultist (Whispers, Madness, Tendril), Leper (Diseased Claw, Contagion, Miasma aura), Darkmage (Shadow Bolt stacking chill), Vampire (Hemorrhage)
Crowd Control	Root: Hunter, Druid, Mage, Cultist, Darkmage; Stun: Knight, Mage, Monk, Prophet; Knockup: Viking (Harpoon), Champion (Berserker Charge); Pull: Viking (Harpoon), Cultist (Open the Gate); Silence: Cultist, Darkmage; Fear: Vampire (Nocturne)
Healing / Support	Druid (Wild Growth, Rejuvenate, Tranquility), Prophet (Bless, Divine Intervention — full HP restore), Chef (Hot Soup, Feast dish buffs), Vampire (lifesteal chain)
Stealth	Vampire (Shadow Step — teleport + 1 s stealth; Nocturne — 6 s invisibility + amplified opener; Mist Step — 180u reposition + stealth), Darkmage (Shadow Cloak — untargetable by ranged)
Execute Mechanics	Champion (Execute doubles damage below 30% HP; Skullsplitter halves CD on kill + 3 bloodtally)
Class Cycling	Master primed classes: Knight, Mage, Monk, Hunter, Druid

Status Effects (26 total)

`slow` · `root` · `stun` · `silence` · `knockback` · `blind` · `taunt` · `shield` · `hot` · `dot` · `lifesteal` · `armor_shred` · `magic_shred` · `untargetable` · `stealth` · `speed_boost` · `damage_boost` ·

`damage_reduction` · `attack_speed_boost` · `bless` · `curse` · `infected` · `chilled` · `revealed` · `fear` · `daze`

Move List by Guild

The full interactive Move List and Guild Dossier are accessible in-game from the main menu. Each dossier shows:

- Full biography
- Stat breakdown (STR / DEX / CON / INT / WIS / CHA each on a 0–20 scale)
- HP, Armor, MR, Move Speed
- Resource name, max, regen rates
- All 6 abilities with combo inputs, damage formulae, cooldowns, and costs
- For Druid: toggle between base and Wolf form ability sets

Below is the complete ability reference for every guild.

Adventurer *(Generalist Bruiser)*

#	Name	Input	Effect
1	Rallying Cry	↓↓J	100u AoE +15% speed/damage to self and allies, 4 s
2	Slash	→→J	80u melee cone, 30 + 0.7×STR
3	Bandage	↓↑J	1.5 s channel self-heal, 50 + 0.5×CON
4	Quickshot	↔J	350u projectile, 35 + 0.6×DEX
5	Adrenaline Rush	↓↑↓↑J	6 s: +40% attack speed, +25% damage, immune to slow/root
RMB	Second Wind	K+J	Restore 30% Stamina

Knight *(Holy Tank)*

#	Name	Input	Effect
1	Holy Rebuke	↓↓J	120u PB AoE, 60 + 1.0×STR holy, stun 1 s
2	Valorous Strike	→→J	60u melee, 40 + 0.8×STR, +10 Resolve on hit
3	Taunt	↓↑J	120u cone, forces enemies to target Knight 3 s
4	Shield Wall	↔J	100u radius, 30% damage reduction to allies 3 s
5	Last Stand	↓↑↓↑J	5 s: cannot drop below 1 HP, +15% damage; ends with 25% self-heal
RMB	Shield Block	K+J	30% damage reduction 2 s

Mage *(Ranged Burst)*

#	Name	Input	Effect
1	Ice Nova	↓↓J	120u PB AoE, 110 + 1.5×INT magical, root 1.5 s
2	Frostbolt	→→J	400u projectile, 80 + 0.8×INT, 30% slow 2 s
3	Blink	↓↑J	Teleport 240u, breaks roots and slows
4	Arcane Shard	↔J	Piercing projectile 500u, 90 + 1.2×INT
5	Meteor	↓↑↓↑J	1.2 s cast, 160u AoE, 200 + 3.0×INT, knockdown
RMB	Short Teleport	K+J	120u step, 20 Mana

Druid *(Healer-Shifter)*

Base Form

#	Name	Input	Effect
1	Wild Growth	↓↓J	100u PB heal circle 5 s HoT, 15 + 0.4×WIS

#	Name	Input	Effect
2	Entangle	→→J	350u projectile, root 2 s
3	Rejuvenate	↓↑J	Heal nearest ally 150u + HoT, 30 + 0.8×WIS
4	Cleanse	↔J	Heal + remove 2 debuffs from ally 150u
5	Tranquility	↓↑↓↑J	Channel 4 s, 200u AoE heal all allies
RMB	Shapeshift	K+J	Toggle Wolf form (5 s CD)

Wolf Form

#	Name	Input	Effect
1	Maul	↓↓J	70u melee, 55 + 0.9×STR, knockdown
2	Charge	→→J	Rush 180u, knockup
3	Roar	↓↑J	150u AoE, 40% slow + fear 1 s
4	Rend	↔J	75u melee + stacking bleed DoT 3 s
5	Primal Fury	↓↑↓↑J	6 s: +40% damage +20% speed
RMB	Revert	K+J	Exit Wolf form

Hunter *(Marksman + Pet)*

#	Name	Input	Effect
1	Disengage	↓↓J	Leap back 150u, blind smoke 1 s
2	Piercing Volley	→→J	3-arrow line 450u, 50 + 0.4×DEX
3	Aimed Shot	↓↑J	500u range, 0.2 s draw, +15% crit chance, 65 + 1.0×DEX
4	Bear Trap	↔J	Ground trap: root 1.5 s + damage on trigger
5	Rain of Arrows	↓↑↓↑J	Channel 3 s, 200u ahead, 6 pulses of 20 + 0.3×DEX
RMB	Pet Command	K+J	Move Wolf pet / cycle its AI mode

Wolf Pet: 200 HP, chaser AI, independently attacks enemies.

Monk *(Melee Assassin)*

Chi is generated by Jab and consumed by other abilities.

#	Name	Input	Effect
1	Serenity	↓↓J	1 s untargetable, cleanse CC, +30% speed 3 s — costs 2 Chi
2	Flying Kick	→→J	Dash 150u, knockup 0.5 s, 40 + 0.8×DEX — costs 1 Chi
3	Jab	↓↑J	50u melee, 25 + 0.6×DEX — **generates 1 Chi**
4	Five-Point Palm	↔J	60u melee + detonation after 4 s (+40 dmg), 110 + 1.2×DEX — costs 3 Chi
5	Dragon's Fury	↓↑↓↑J	Channel 2 s, 5 strikes, final stun 1.5 s — costs 5 Chi
RMB	Parry	K+J	0.3 s parry window; success = stun

Viking *(Berserker)*

Rage builds from melee combat and decays out of combat.

#	Name	Input	Effect
1	Whirlwind	↓↓J	Channel 2 s, 100u PB, 25 + 0.5×STR, 10% lifesteal
2	Harpoon	→→J	400u projectile, pulls target 120u, 40 + 0.7×STR
3	Bloodlust	↓↑J	5 s: +25% attack speed, 10% lifesteal
4	Axe Swing	↔J	60u melee cone, 35 + 0.8×STR, builds Rage
5	Undying Rage	↓↑↓↑J	3 s: cannot die; damage taken converts to 30% heal on expiry
RMB	Shield Bash	K+J	Knockback 150u, 40 + 0.5×STR

Prophet *(Cleric / Buffer)*

#	Name	Input	Effect
1	Prophetic Shield	↓↓J	Shield 120 + 1.2×WIS; on break, heals 30 + 0.5×WIS
2	Smite	→→J	450u holy projectile, 70 + 0.9×WIS, reveals target 3 s
3	Bless	↓↑J	Ally 200u: +15% damage +10% speed 8 s
4	Curse	↔J	Target 300u: +20% damage taken, -15% damage dealt 6 s
5	Divine Intervention	↓↑↓↑J	Target 200u: invulnerable 3 s, heals to full on expiry
RMB	Divine Insight	K+J	Reveal all enemies in 360u for 4 s

Vampire *(Stalker Lifesteal)*

#	Name	Input	Effect
1	Hemorrhage	↓↓J	300u projectile, 5 s DoT; overflow heals Vampire
2	Shadow Step	→→J	Teleport behind nearest enemy + 1 s stealth
3	Blood Drain	↓↑J	Channel 2 s, 70 + 1.0×DEX, heals full damage dealt
4	Fang Strike	↔J	60u melee, 30 + 0.7×DEX, heals 60% of damage
5	Nocturne	↓↑↓↑J	6 s: invisible, +50% speed; next ability from stealth +100% damage + fear
RMB	Mist Step	K+J	180u reposition + 1 s stealth

Cultist *(DoT Caster)*

Sanity is inverse: abilities have *negative* cost (casting gains Sanity). Sanity decays over time, creating a pressure loop.

#	Name	Input	Effect
1	Summon Spawn	↓↓J	Summon Drowned Spawn (150 HP, 20 s)
2	Whispers	→→J	400u psychic projectile, silence 1 s, 30 + 0.8×INT
3	Madness	↓↑J	120u AoE, 0.8 s confusion + DoT, 30 + 0.6×INT
4	Tendril Grasp	↔J	Ground tendril 100u, root + DoT 4 s
5	Open the Gate	↓↑↓↑J	Channel 3 s, 240u AoE, pulls all to center, 150 + 2.0×INT
RMB	Gaze into Abyss	K+J	Next ability: 0 Sanity cost + 30% damage

Champion *(Forward-Only Bruiser)*

Bloodtally stacks from kills and certain abilities. Execute doubles vs targets below 30% HP.

#	Name	Input	Effect
1	Tithe of Blood	↓↓J	Consume 3 stacks: +30% attack speed 5 s + self-heal
2	Berserker Charge	→→J	Dash 240u, knockup 0.5 s, +1 Bloodtally, 35 + 0.6×STR
3	Execute	↓↑J	60u melee; ×2 damage if target <30% HP
4	Cleaver	↔J	60u melee arc, 40 + 0.9×STR
5	Skullsplitter	↓↑↓↑J	Heavy melee; on kill: halve CD + 3 Bloodtally
RMB	Challenge	K+J	Target 300u takes +20% damage from Champion 5 s

Darkmage *(Shadow Controller)*

#	Name	Input	Effect
1	Darkness	↓↓J	90u vision-reduction zone 4 s
2	Grasping Shadow	→→J	400u projectile, root 1.5 s, 30 + 0.6×INT
3	Soul Leech	↓↑J	360u target, 60 + 1.0×INT, restores 50% damage as Mana

#	Name	Input	Effect
4	Shadow Bolt	↔J	400u projectile, stacking chill −5% speed (max 5 stacks), 45 + 0.9×INT
5	Eternal Night	↓↑↓↑J	240u silence zone + shadow DoT 6 s
RMB	Shadow Cloak	K+J	2 s: +60% speed, untargetable by ranged

Chef *(Utility Support)*

Dishes are consumable buffs. Feast distributes them to allies. Signature Dish combines them.

#	Name	Input	Effect
1	Feast	↓↓J	Apply dish buff to all allies in 180u
2	Ladle Bash	→→J	60u melee, daze 0.5 s, 40 + 0.5×STR
3	Hot Soup	↓↑J	Heal + regen 20/s × 4 s to ally 240u
4	Spice Toss	↔J	300u projectile, blind 3 s + DoT, 45 + 0.2×INT
5	Signature Dish	↓↑↓↑J	Channel 2 s; combined dish effects on all allies 300u for 20 s
RMB	Pocket Dish	K+J	Consume a dish for its buff (self)

Leper *(Diseased Bruiser)*

Rot builds over time. Miasma toggles an aura that constantly damages nearby enemies.

#	Name	Input	Effect
1	Plague Vomit	↓↓J	80u cone, infected DoT + 30% slow 2 s, 40 + 0.6×STR
2	Diseased Claw	→→J	60u melee + necrotic DoT 3 s, 18 + 0.7×STR
3	Necrotic Embrace	↓↑J	Grab: heals 50% of damage dealt, 40 + 0.8×STR
4	Contagion	↔J	Infect target 300u for 4 s

#	Name	Input	Effect
5	Rotting Tide	↓↑↓↑J	Channel 3 s, 180u PB; enemies killed revive as Rotting Husks (5 s)
RMB	Miasma	K+J	Toggle aura: (2 + 0.1×CON) necrotic DPS in 90u radius

Master *(Prestige Hybrid)*

Primed classes: Knight · Mage · Monk · Hunter · Druid. Class Swap (RMB) cycles them. Eclipse rotates abilities through all five variants for 5 s.

#	Name	Input	Effect
1	Chosen Strike	↓↓J	Ranged or melee depending on primed class, 40 + 1.0×STR
2	Chosen Utility	→→J	Utility from primed class (default: teleport)
3	Chosen Nuke	↓↑J	Burst from primed class +10%, 120 + 1.1×INT
4	Eclipse	↔J	5 s: ability slots cycle through all five primed-class versions
5	Apotheosis	↓↑↓↑J	10 s: all CDs halved, +20% damage, +2% HP/s
RMB	Class Swap	K+J	Cycle primed class (1 s CD)

The Nine Stages

Each stage has its own colour theme, atmospheric flavour text, and background treatment. All nine are selectable in VS and Multiplayer modes.

Stage	Colour	Flavour
Assembly Hall	Blue (hue 210)	"Flagstones under torchlight. The Knights swore here — mind the pillars."
Night Market	Amber (hue 40)	"Paper lanterns and slick cobbles. The stalls are open. The knives, also."

Stage	Colour	Flavour
Rot-Kitchen	Green (hue 95)	"The Lepers took the stoves. The stew is old and moves on its own."
Mage Tower	Purple (hue 260)	"Levitating glass floors. Do not step where the runes are singing."
Moonwake Grove	Teal (hue 140)	"The old trees listen. Tread polite. The Druids are watching."
Drowned Catacombs	Magenta (hue 300)	"Stalactites, saltwater, and names written in a dead vowel."
Red Throne	Red (hue 15)	"The throne counts. Every retreat is a step into the pit."
Vampire Docks	Rose (hue 330)	"Fog off the pier. Nothing docks here that returns."
Monastery Rooftops	Cyan (hue 185)	"Slate tiles, thin air, one misstep to the courtyard below."

Story Mode

A single-player left-to-right run across the Plains of Nan. Waves spawn ahead of you as you advance. Die once and the run ends.

Wave Progression

Wave	Trigger X	Enemies
1	400	2× Plains Bandit
2	900	3× Plains Bandit + 1× Bandit Archer
3	1600	2× Wolf
4	2400	2× Plains Bandit + 2× Bandit Archer
5	3200	1× Bandit Brute + 2× Wolf
6 (Boss)	3800	**Bandit King**

Enemy Roster

Enemy	HP	Notes
Plains Bandit	160	Melee chaser, armor 5
Bandit Archer	120	Ranged, 300u projectile range
Wolf	120	Pack AI, fast
Bandit Brute	320	Slow (70 speed), heavy armor 15, hits for 18
Bandit King	3200	Boss. Armor 20, MR 10, 35 damage, 80u range, 1000ms attack CD

Summons (player guild abilities)

Summon	HP	Source
Wolf Pet	200	Hunter
Drowned Spawn	150	Cultist
Rotting Husk	200	Leper (Rotting Tide on-kill)

VS & Multiplayer

Match Flow

```
lobby → char_select → stage_select → loading → in_game → results
```

- **Rounds:** Best-of-3 (win 2 rounds to claim the match)
- **Round win condition:** Last player standing
- **Multiplayer:** Server authoritative — clients send inputs only, receive state
- **Room system:** Private by default; share room code to invite. Rooms can be set public.
- **Lobby features:** Ready toggle (R), in-lobby chat (100 message history), room code display
- **Rematch:** Available from results screen (Enter)

Results Screen

After each match the game displays a full breakdown for both players:

Stat	Description
Damage Dealt	Total damage output
Damage Taken	Total damage received
Abilities Cast	Number of ability activations
Max Combo	Highest consecutive hit streak
Crit %	Percentage of hits that were critical strikes
Healing	Total healing done

Winner side is highlighted; loser is greyed. Guild tags are displayed for both players (e.g., `HOLY TANK`, `STALKER LIFESTEAL`).

The HUD

Top Bar

- **Player 1 slot:** Guild monogram · health bar (red) · resource bar (guild colour) · resource label
- **Centre:** Stage name · round timer (VS/MP only)
- **Player 2 / Boss slot:** Mirror of P1 (VS), or Boss name + oversized health bar (Story)

Ability Strip

Six cards along the bottom — five combo abilities plus the RMB utility. Each card shows:

- Emoji icon
- Ability name
- Cooldown timer overlay (if on cooldown)
- Keybind label (1–5, RMB)

Druid shapeshifted into Wolf form updates all six cards to Wolf abilities. Master updates cards when the primed class changes.

Combat Log

Scrolling message history (100-message cap) showing hits, abilities, status effects, and kills in real time.

Round Indicators

Dots or pip icons tracking round wins (0–3) per player, visible below the health bars.

Stats & Resources

The Six Attributes

Each guild has scores in STR / DEX / CON / INT / WIS / CHA (scale 0–20). These scale ability damage via formulae like `35 + 0.8×STR` or `80 + 0.8×INT`.

Defences

- **Armor** — reduces incoming physical damage. Range: 4 (Cultist, Darkmage) to 12 (Knight, Champion).
- **Magic Resist** — reduces incoming magical/holy/shadow damage. Range: 5 (Adventurer, Hunter, Viking, Champion) to 20 (Mage, Prophet).

Resource Systems

Every guild runs a unique resource that shapes its playstyle:

Resource	Guild	Max	Behaviour
Stamina	Adventurer, Chef	100	+5 idle / +2 in combat
Resolve	Knight	100	No regen; -1 decay; built by Valorous Strike
Mana	Mage, Darkmage	200	+5 idle / +2 in combat (Darkmage +4 idle)
Essence	Druid	100	+4 idle / +2 in combat
Focus	Hunter	100	+3 idle / +3 in combat
Chi	Monk	5	No regen; generated by Jab; consumed by abilities
Rage	Viking	100	Starts at 0; built by Axe Swing; -2 decay
Faith	Prophet	100	+3 idle / +2 in combat

Resource	Guild	Max	Behaviour
Bloodpool	Vampire	100	Starts at 50; -1 decay
Sanity	Cultist	100	Starts at 0; -2 decay; abilities have negative cost (casting gains Sanity)
Bloodtally	Champion	10	Stacks from kills and specific abilities; no decay
Rot	Leper	100	Starts at 0; +3 idle / +3 in combat
Mastery	Master	200	+5 idle / +3 in combat

Audio System

No audio files exist in this project. Everything you hear is synthesised in real time via the **Web Audio API** (`src/audio/audioManager.ts`).

Sound Effects

Event	Synthesis
Attack	Sawtooth + square (220 Hz + 180 Hz)
Critical Hit	Twin square tones (440 Hz + 660 Hz)
Heal	Rising sine chord (523 · 659 · 784 Hz)
Block	Deep square (150 Hz)
Parry	Sharp twin square (880 Hz + 1100 Hz)
Jump	Frequency sweep 200→400 Hz
Land	Low square thump (100 Hz)
Knockdown	Deep square drop (80 Hz)
Death	Descending sawtooth 440→55 Hz over 0.5 s
Ability Cast	Sine chord (660 Hz + 880 Hz)
Impact	Sawtooth punch (120 Hz)
Insufficient Resource	Descending double tone (200 Hz + 150 Hz)

Event	Synthesis
Victory	Rising arpeggio (523 · 659 · 784 · 1047 Hz, 120 ms apart)
Defeat	Falling arpeggio (523 · 494 · 440 · 392 · 349 Hz, 200 ms apart)

Music

- **Stage music:** 12-note square wave loop at 350 ms/note
- **Boss music:** Darker 12-note sawtooth loop at 350 ms/note, lower register
- Music fades in/out on phase transitions
- Master volume, music gain, and SFX gain are all independently adjustable and persisted to localStorage

Balance Tooling

One of the most unusual features of Nannymud is a fully automated headless balance runner.

What It Does

`scripts/balance-runner.ts` runs **4,500 bot-vs-bot matches** — every possible guild pairing (225), 20 matches each — and produces a 15×15 win-rate matrix.

```
npx tsx scripts/balance-runner.ts
# runs ~3–5 minutes
# prints matrix to stdout
# writes scripts/balance-output.csv
```

How It Works

- Ticks at 16 ms (real-time equivalent)
- CPU difficulty: maximum
- Stage: fixed (Assembly Hall)
- Alternates sides each iteration to cancel spawn-position bias
- Each match has a 120 s timeout
- Seeded RNG per match (deterministic runs)

Tuning Levers

All changes are made in `packages/shared/src/simulation/guildData.ts`:

1. **Strategy block** — `priority`, `useAtCloseRange`, `useAtLongRange`, `retreatBelowHpPct`, `blockOnReaction`, `preferRange` — adjusts AI behaviour only, no effect on human play
2. **Stats** — `hpMax`, `armor`, `magicResist`, `moveSpeed`
3. **Ability params** — `baseDamage`, `cost`, `cooldownMs`

Target: all 15 guilds within 40–60% win rate across all matchups. The current committed baseline is at `scripts/balance-output.csv`.

Autonomous Build Agents

Nannymud is also a platform for an autonomous multi-agent development system. Agents run continuously from a GitHub Issues queue and build the game while you sleep.

Agent Roles

Agent	File	Role
Orchestrator	`agents/orchestrator.md`	Reads open issues, assigns lanes, dispatches sub-agents, moves labels (todo → in-progress → qa → done), posts results
Dev	`agents/dev.md`	Executes one bounded task per git worktree; runs typecheck + build + tests; posts completion to issue
QA	`agents/qa.md`	Verifies dev work, runs manual test scope per task, posts pass/fail
Reviewer	`agents/reviewer.md`	Code review and feel-sensitive feedback on design/balance; gates merge decisions
Asset	`agents/asset.md`	Generates pixel art, animations, and tilesets via PixelLab

Label Taxonomy

Issues flow through state via GitHub labels:

```
todo → in-progress → qa → done
                   ↘ todo (on QA fail, with rejection comment)
```

Lane labels: `lane:dev` · `lane:asset` · `lane:design` · `lane:qa` · `lane:reviewer`

Running the Orchestrator

```
orchestrator run
```

This starts a self-paced `/loop`. The orchestrator reads open `todo`-labelled issues, dispatches at most 1 dev + 1 design + 1 asset agent simultaneously (disjoint file scopes), and advances labels as work completes. GitHub token must be set via `GITHUB_TOKEN`.

Tech Stack

Layer	Technology
Dev server & bundler	Vite 5
UI / menus	React 18
Game engine	Phaser 3
Language	TypeScript (strict) — client, shared, server
Multiplayer	Colyseus (server-authoritative, @colyseus/schema state sync)
Audio	Web Audio API — 100% procedural, no audio files
Sprites	PixelLab AI (80 px, displayed at ×1.5 via DISPLAY_SCALE)
Monorepo	npm workspaces (`packages/shared`, `packages/server`)
Tests	Vitest + a golden determinism snapshot for MP correctness
Linting	ESLint flat config with custom rule: no browser APIs in simulation
Simulation	Pure TypeScript, deterministic, shared client/server

Architecture Principle

The simulation layer (`packages/shared/src/simulation/**`) is pure — no Phaser, no DOM, no `Math.random()` , no `Date.now()` . This is enforced by ESLint and gated by the golden determinism test. The same simulation runs identically in the browser (single-player) and on the Colyseus server (multiplayer authority).

Built with Vite · React · Phaser 3 · Colyseus · Web Audio API · PixelLab · TypeScript